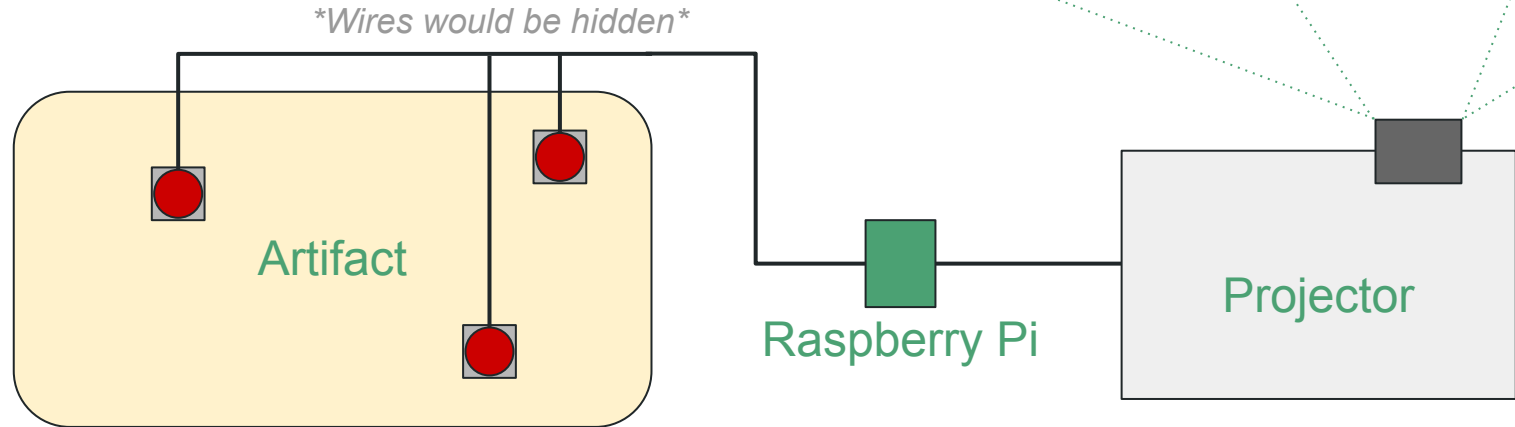


Wall Projections

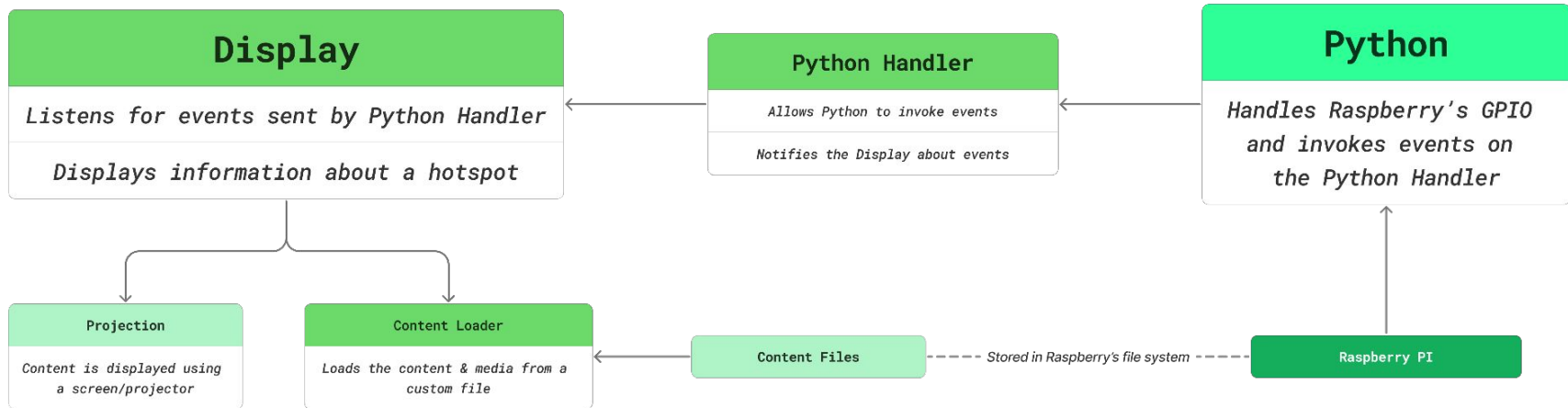
Interactive Exhibitions for Bristol Museum

Our MVP

An artifact is set up with buttons placed at points of interest. When a user presses a button, information related to the point of interest is displayed on the Projector.

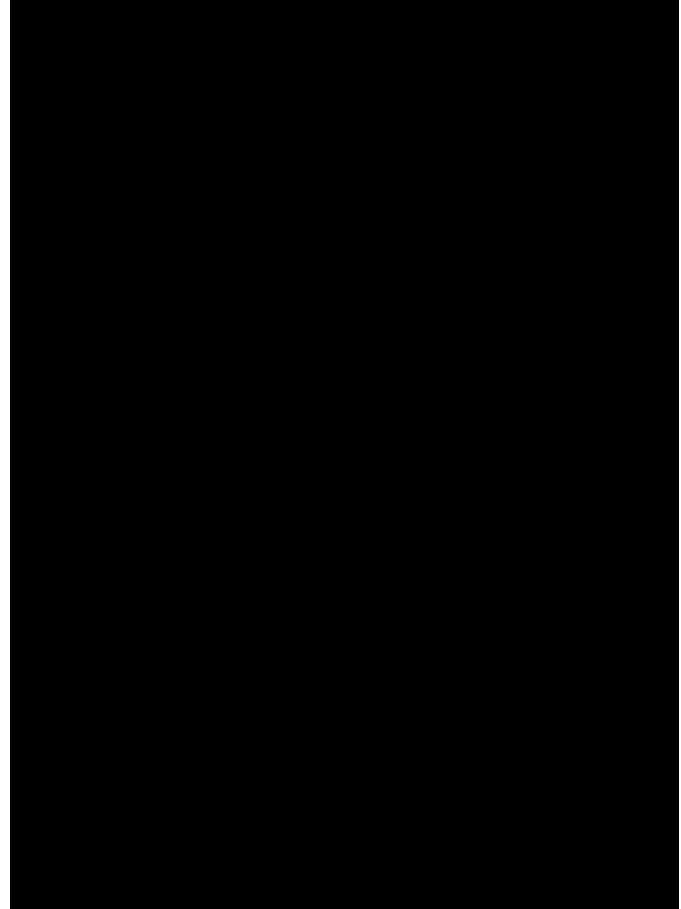
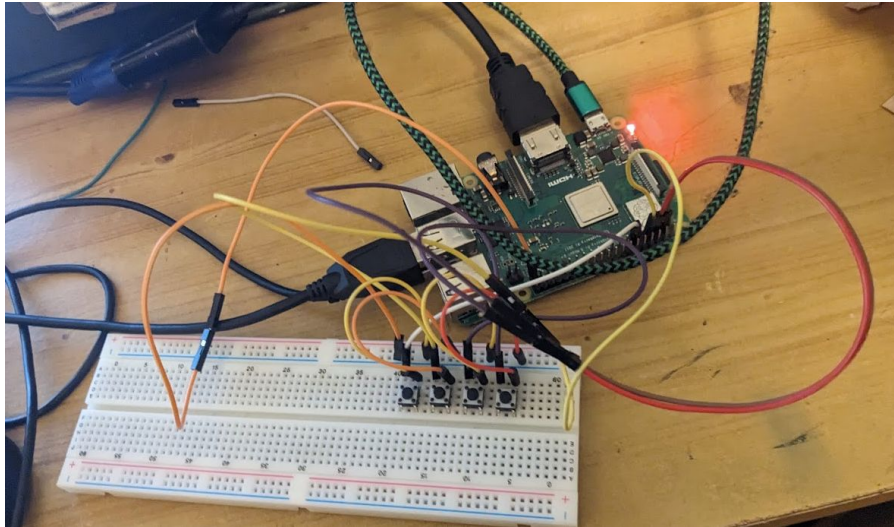


High-level System Architecture



Demo

Include fallback images/video



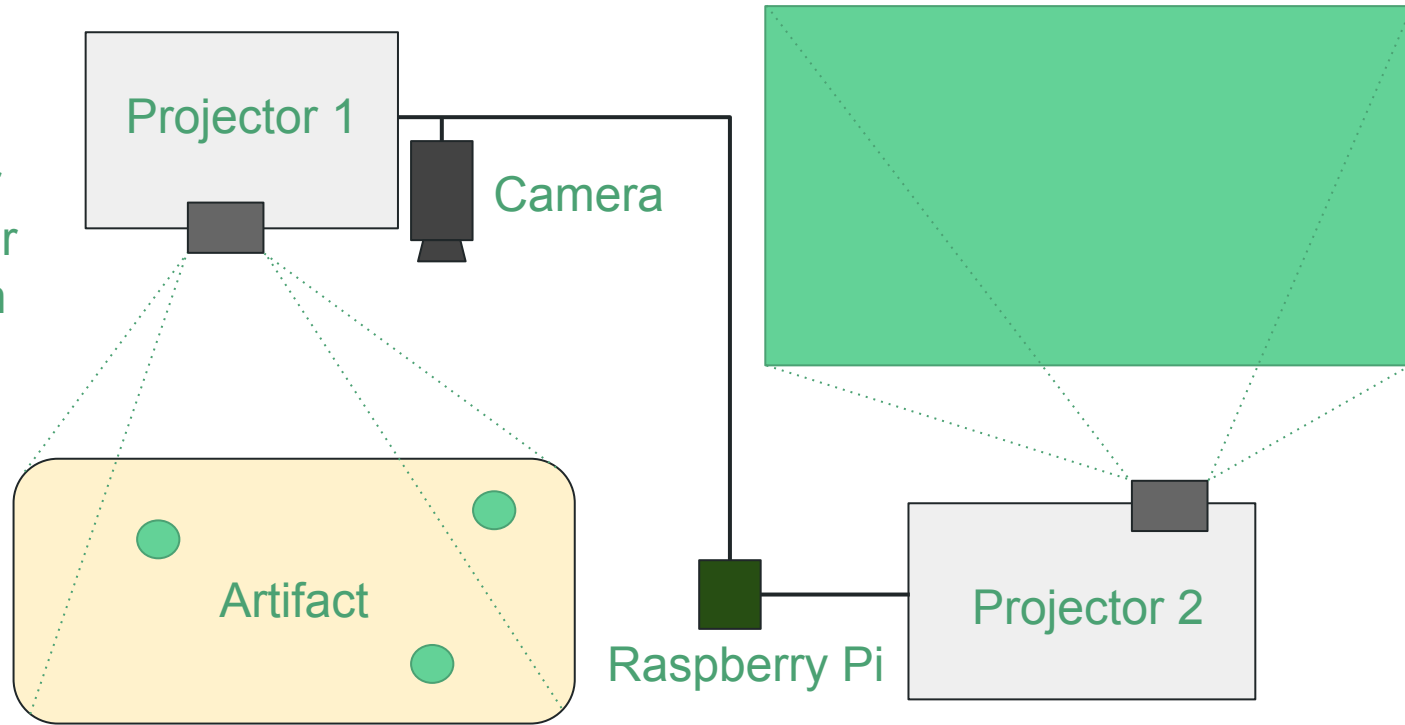
Technology Stack

Our project is written in 2 languages: *C#* for the Frontend and *Python* for the Backend. The former is built using *Avalonia*, a UI framework for *.Net* platform, and *python.net* library to communicate with the Backend. Currently, the Python component takes Raspberry Pi GPIO as input, but later on it will be replaced by Google's MediaPipe.



Possible final design

An artifact is set up with hotspots projected at points of interest. When a user places their hand over a hotspot, information related to the hotspot is displayed on Projector 2.



End goal

- We would like to implement an Editor meaning the Bristol Museum can easily add and remove points of interest and customize the information associated with them
- We need to further develop hand tracking and how we want to display hotspots
- We want to implement an easy and intuitive method of calibrating the whole system, so that the interactive artifacts can be set up quickly

Initial Research & Challenges

- Raspberry Pi setup & GPIO
- UI libraries
 - JavaFX
 - Cross-compilation issues
 - Video View issues
- Computer Vision
- JSON (De)serialisation



Questions